A One-Class Classification Model for Burned-Area Detection Based on Mutual Ordering of Normalized Differences

Massimo Zanetti, *Member, IEEE*

***Abstract*— Global-scale assessment of burned area (BA) is essential for climate studies and open access satellite-borne multispectral (MS) imagery is vital for the mapping purpose. Global characterization of BAs via MS analysis is difficult as it usually requires ancillary data to model local factors such as vegetation types and local ecoclimate systems. This article proposes a novel classification model that exploits certain mathematical properties of normalized difference indexes (NDIs) to build an abstract space of features where the BA class can be learned globally and solely using MS images. The core idea is that, although NDIs are subject to strong intraclass variations, their mutual order (i.e., the sign of their difference) can be robust enough for characterization. By encoding every possible such ordering relation in a binary domain, the feature space turns out to be hyperdimensional, with abstraction capabilities similar to that of neural network (NN) layers. The proposed classification model is one-class and, therefore, is very convenient as it only requires training samples for the positive class to be collected. The model is experimentally validated in an extensive BA detection exercise with Sentinel-2 images that involves recently published global BA reference data. Results are promising as we report higher $F1$-score than those reported for state-of-the-art BA products currently available, which are obtained through hybrid techniques and multiple data sources. The model also outperforms the well-known and largely used one-class support vector machine (OC-SVM), which is tested in this work for the first time for BA detection at a global scale.**

***Index Terms*— Hyperdimensional binary spaces, normalized difference indexes (NDIs), one-class classification, remote sensing (RS).**

I. Introduction

THE increasing awareness about the climate change problem highlights more than ever the central role that remote sensing (RS) plays in providing global-scale assessment of many environmental and anthropic phenomena [1], [2]. The exploitation of RS spaceborne multispectral (MS) imagery is essential to many Earth mapping tasks because of the spectral content available in images, which enables the recognition of many land-cover classes and their changes due to full land coverage with repeating pass schemes granting new images every few days [3]. The open access policy associated with the major MS data missions, such as Sentinel-2 (S2), from the European Space Agency (ESA), and Landsat-8/9 (L8/9), from the United States Geological Survey (USGS), brings enormous amounts of data at disposal and makes automatic analysis vital for uptaking the challenge. As recognized by the Global Climate Observing System (GCOS), fire disturbance is an essential climate variable (ECV) for the understanding of several key environmental processes such as vegetation dynamics, desertification and soil erosion, hydrological systems, and atmospheric emissions, [4], [5]. Burned area (BA) refers to the total area affected by a fire and is considered the primary descriptor of the fire disturbance, being itself the most relevant factor for the estimation of fire causes and effects [6].Burning biomass produces different spectral signals observable from space: direct radiation (heat from the fire front), aerosols and smoke, char and ash residuals, and scarred vegetation [7]. BA analysis is based on the estimation of solid residuals and vegetation scar through their impact on spectral measurements [8]. Charcoal residue and vegetation scar are responsible for quite specific spectral signatures recognizable in MS images [7], [9]. BA detection approaches typically try to detect them using locally adaptive or physical-driven methods performing discrimination over a set of attributes (reflectance bands or spectral indices) from which discriminant functions are created by maximizing interclass and minimizing intraclass variation [10]. Examples are Bayesian classifiers [11], random forests [12], and support vector machines (SVMs) [13]. Although we cannot go through the whole BA detection-related literature (see [10] for an excellent covering of the topic), it is widely agreed that the approaches proposed in these studies are valuable for obtaining BA detailed information over limited areas [14], [15], [16], while they fail in performing the same when applied at larger scales [17], [18]. The main limiting factors are essentially two intrinsic properties of the BA class, which is both: 1) spectrally heterogeneous, with signatures varying significantly according to different local factors such as vegetation type and density, soil wetness/dryness, local climate conditions, and signal persistence over time [19], [20], and 2) rare, considering the typical reduced spatial proportion of fire patterns extent with respect to whole MS scenes [17]. This is the reason why the common approach in BA detection methods aiming at global-scale production is including ancillary information to provide classifiers with supervised knowledge about: 1) vegetation strata and/or biomes maps and 2) active fire (AF) locations, spotted by anomalies in thermal bands. The former is to make the spectral representation of the classifiers locally adaptive, and the latter is to provide reliable spatial localization

Manuscript received 22 June 2023; revised 11 July 2023 and 24 July 2023; accepted 30 July 2023. Date of publication 2 August 2023; date of current version 16 August 2023.

The author is with the Center for Digital Society, Fondazione Bruno Kessler, 38123 Trento, Italy (e-mail: mazanetti@fbk.eu).

Digital Object Identifier 10.1109/TGRS.2023.3301056

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

of certain BA signals. Although at the operational level, these approaches provide pragmatic solutions for production, no shared theoretically founded model exists in the literature that enables the representation of the BA class in a consistent manner all around the globe, and the even more pressing question remains open of whether such a model could exist being only dependent on MS information.

This article gives a contribution in this direction, by proposing a one-class classification model that transforms the MS domain in such a way that the BA class can be robustly represented. One-class classification provides an inference model for one single target class (the so-called positive class) and is different from binary classification in that it does not require an explicit model for its complementary class (also called the negative class) [21]. This is a convenient approach for detection of low prior probability events in large sets of images because it prevents disproportionate efforts in modeling the negative class, which is in the RS case typically vast and diverse, therefore too complex to be effectively modeled [22]. BAs are, from an RS perspective, rare events both in space and time, and therefore, they fit very well into this design. One-class classification approaches can be essentially categorized as boundary methods, where bounds are found in different domains such as [23]: linearly transformed spaces [e.g., principal component analysis (PCA)] [24], probability density spaces [25], logistic regression [21], and kernel spaces above all, with the successful and largely used one-class SVM (OC-SVM) typically used for outlier detection [26], [27]. Note that these methods are specializations of their standard classification counterparts, with the only difference that actual boundaries are not induced by competition between classes but rather by predetermined hyperparametric choices and constraints. A few attempts were made in the literature to apply the one-class classification concept to BA detection using, for example, PCA [28], logistic regression [29], and OC-SVM [30]. However, it turns out that the standard domain transformation techniques mentioned above cannot provide appropriate translation of the MS domain such that the BA class can be enclosed by trivial boundaries, and mitigation techniques are often required, such as modeling the unburned class as well, to improve the accuracy of detection over large extents [13].

This article proposes an entirely novel transformation function that enables a hyperdimensional representation of the MS domain in which the BA class can be modeled by a trivial subspace. The proposed transformation of the MS domain is obtained from the formal generalization of some empirical relationships expressed by normalized difference indexes (NDIs) when measured over BAs, and it provides a very abstract description of the data that are similar to that of typical neural network (NN) layers, even if substantially different, as its intrinsic coordinates turn out to be highly nonlinear. In brief terms, each coordinate encodes in binary value the outcome of the mutual comparison between two NDIs. The resulting feature space is hyperdimensional as the number of such possible comparisons increases exponentially with the number of spectral bands in input. The proposed model is supported by a formal definition of learnability conditions that a set of positive samples must fulfill for it to be used for training. We can summarize the main contributions of this work in the following points.

1) The method is one-class and solely relying on MS images, and therefore, it provides theoretical insides on the spectral properties of BAs. Moreover, such a deep use of NDIs, given their intrinsic connection with vegetation-related physical processes, gives the proposed model the ability to be potentially tailored around vegetation-related target classes other than BAs.
2) The proposed model generalizes an intuitive concept making the resulting hyperdimensional representation transparent and with a definite and precise meaning. This is in contrast to most applications of NNs, which rely on deep-layer embeddings that are very dependent on users' choices and typically do not provide knowledge-based explanations of the observed classes.
3) For the first time, BA detection is conducted in a nearly operational scenario (several images all around the globe) by using a nonhybrid classification procedure.
4) The method outperforms the accuracy of baseline BA products and state-of-the-art one-class methods.

The rest of this article presents in detail the rationale that led to the idea at the core of the proposed model in Section II, its formal treatment and algorithm implementation in Section III, and its validation in Section IV. Finally, Section V draws the conclusions and presents perspectives for further extension of the study presented in this work.

II. Rationale

This section introduces normalized difference indices and provides the core intuitive idea for the proposed one-class classification model, which is formally addressed in Section III.

A. Notation

Let us denote an MS image acquired at time t as a collection $\boldsymbol{X}^t$ of spectral values $\boldsymbol{x}^t = (x_1^t, \ldots, x_B^t)^T \in \mathbb{R}_+^B$, where B is the number of spectral bands. We will sometimes refer to specific spectral bands by their common abbreviated nomenclature (e.g., x_{RED}, x_{NIR}, and x_{SWIR}) when emphasizing their portion in the spectrum is more important than their indexing in the coordinate system. Given two MS images acquired at times t_1 and t_2, we obtain the compound image obtained from $\boldsymbol{X}^{t_1}$ and $\boldsymbol{X}^{t_2}$ by concatenation of their spectral values $\boldsymbol{x} = (x_1^{t_1}, \ldots, x_B^{t_1}, x_1^{t_2}, \ldots, x_B^{t_2})^T \in \mathbb{R}_+^{2B}$ [31]. For convenience, the compound MS variable $\boldsymbol{x}$ is notated throughout this article in a simplified manner as $\boldsymbol{x} = (x_1, \ldots, x_N)^T \in \mathbb{R}_+^N$, where $N = 2B$ (spectral indexing is joined) and temporal indexing is dropped.

B. Normalized Difference Indexes

NDIs are ubiquitous and predominant in (but not limited to) vegetation-related studies because of their success in explaining several plausible correlations between spectral reflectance and some physical parameters descriptive of vegetation characteristics. The normalized difference between two measured

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

positive[1] spectral values $x_a, x_b > 0$ is given by the algebraic (band-math) expression

$$x_a|x_b := \frac{x_a - x_b}{x_a + x_b} \tag{1}$$

which is bounded to return values in the range $(-1, 1)$. This formulation highlights the interaction between an abundance and an absorption component (x_a and x_b, respectively) reaching values close to 1 when the observed abundance and absorption are at their maximum peaks simultaneously [32]. For different pairs of spectral bands in input, NDIs can provide various descriptions of vegetation-related phenomena. Among many combinations proposed in the RS literature, let us recall some of them that are deeply connected with the study of BAs and also appropriate for explaining the rationale behind the proposed model. The normalized difference vegetation index (NDVI) emphasizes the RS signal of healthy vegetation. Leaf cells in living green plants absorb much of the solar radiation in the visible spectrum reaching the maximum in the red (RED) portion and emit in the near-infrared (NIR) spectral region as a result of the photosynthetic process [33], [34]. In the NDVI formulation

$$\mathrm{NDVI}^t = \frac{x^t_{\mathrm{NIR}} - x^t_{\mathrm{RED}}}{x^t_{\mathrm{NIR}} + x^t_{\mathrm{RED}}} = x^t_{\mathrm{NIR}}|x^t_{\mathrm{RED}} \tag{2}$$

values close to 1 suggest for healthy vegetation, whereas other values indicate absent or damaged vegetation conditions, including burn scars. An enhanced and more specific description of the RS signal of BAs is possible due to the characterization of various postfire effects, such as exposure of bare soil, presence of charcoal or ash, and warm temperatures. The normalized burned ratio (NBR) enhances the description of areas affected by a fire as they typically show low response both in the visible range of the spectrum, due to the presence of dark matter, and in the NIR region because of the interrupted photosynthetic activity (as opposed to the behavior of healthy vegetation), and stronger signal in the shortwave infrared (SWIR) region due to higher temperatures and the low evapotranspiration [35]. In the NBR formulation [36]

$$\mathrm{NBR}^t = \frac{x^t_{\mathrm{SWIR}} - x^t_{\mathrm{NIR}}}{x^t_{\mathrm{SWIR}} + x^t_{\mathrm{NIR}}} = x^t_{\mathrm{SWIR}}|x^t_{\mathrm{NIR}} \tag{3}$$

values close to 1 suggest for burned vegetation or exposed dark bare areas. The modified normalized burned ratio (NBR2) formulation is an alternative version of NBR that enhances sensitivity to water in vegetation when different measurements in the shortwave infrared are available. In the NBR2 formulation [37]

$$\mathrm{NBR2}^t = \frac{x^t_{\mathrm{SWIR2}} - x^t_{\mathrm{SWIR1}}}{x^t_{\mathrm{SWIR2}} + x^t_{\mathrm{SWIR1}}} = x^t_{\mathrm{SWIR2}}|x^t_{\mathrm{SWIR1}} \tag{4}$$

we have SWIR2 > SWIR1 in terms of wavelength.

C. Proposed Concept

Despite the compelling rationale underlying NDIs derivation, which seems to capture the main spectral evidence of fire effects from different viewpoints, fire impacts on vegetation are neither binary nor homogeneous in terms of spectral representation [38], [39]. Although NDIs are in many cases well correlated to fire severity estimates based on biomass loss [14], [40], their values are nonetheless strongly dependent on local ecosystems, topography, vegetation healthiness, density and spatial distribution, weather conditions, and other key factors [20], [41], [42]. They can vary considerably depending on specific conditions local to the scene and they cannot provide an absolute measurement of effects related to the burning process [43]. For this reason, most of the BA literature focuses on change detection approaches instead of single postdisturbance analysis, the latter usually requiring unrealistic assumptions about the homogeneity of the predisturbance landscape [44]. Change detection approaches aim at characterizing burn effects through quantification of observable variations, such as albedo decrements, augmented temperatures, biomass loss, interrupted photosynthetic activity, sparse vegetation, and decreased soil moisture. These variations can be measured in relation to the physical meaning of NDIs by the difference between postfire and prefire observations. Temporal differences such as [45], [46]

$$\begin{aligned} \mathrm{dNBR} &= \mathrm{NBR}^{t_2} - \mathrm{NBR}^{t_1} \\ \mathrm{dNDVI} &= \mathrm{NDVI}^{t_2} - \mathrm{NDVI}^{t_1} \end{aligned} \tag{5}$$

where t_1 and t_2 are prefire and postfire dates, respectively, bring valuable spectral information about the change induced by fire. However, global studies on such fire-induced spectral changes emphasized their ambiguous and somewhat contradictory nature, highlighting in particular how the change characterization still remains essentially locally driven [10], [18], [38]. It is the case of apparently contradictory outcomes, for instance, when prefire conditions consist of sparse vegetation on bright soils [46], leading to albedo increase after burning, or in boreal forests, where the result of burning is melted permafrost, which corresponds to increased soil moisture [47].

Current one-class classification approaches cannot perform well on global characterization of BA being their associated transformation techniques not capable of projecting dNBR, dNDVI values (and spectral indices in general) into trivial boundaries. The main assumption behind our model is that, even if actual values of NDIs are subject to strong intraclass variations, the sign of their difference (i.e., the binary increase/decrease information) is more likely preserved within the class. For the sake of explanation, let us assume that conditions like

$$\begin{aligned} \mathrm{NBR}^{t_2} - \mathrm{NBR}^{t_1} &> 0 \\ \mathrm{NDVI}^{t_2} - \mathrm{NDVI}^{t_1} &< 0 \end{aligned} \tag{6}$$

are empirically verified over BAs. This reflects the expectation that burned signal augments and vegetation signal decreases, as a result of fire action. It is obvious that the above two relations can only be necessary conditions for the characterization of Bas, and certainly, they can not be sufficient. The natural question that follows is whether we can reach a sufficient set of conditions by testing the sign of all NDI differences that is

[1]The assumption that spectral measurements cannot take null values does not constitute a limitation in practical terms.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

possible to make, i.e., verifying inequalities

$$\frac{x_i - x_k}{x_i + x_k} - \frac{x_h - x_j}{x_h + x_j} \lessgtr 0 \tag{7}$$

for any possible meaningful choice of x_i, x_j, x_k, and x_h in the compound MS domain. Given the number of spectral bands available in MS images, the resulting feature space turns out to be very high dimensional and the set of verified conditions corresponds to a specific subspace. In the remainder of this article, this decision model is developed formally and tested for the characterization of the BA class.

III. One-Class Classification Model

This section provides the theoretical and implementation details for the model intuitively introduced and outlined in Section II.

A. Mutual Ordering of Normalized Differences Transformation

The main point in this section is to define the mathematical transformation that turns the MS domain into a binary space that encodes the sign of every possible pairwise NDI difference. We, therefore, look for a map

$$\begin{aligned} \varphi : \mathbb{R}_+^N &\to \{0, 1\}^Q \\ \boldsymbol{x} &\mapsto \boldsymbol{z} \end{aligned} \tag{8}$$

where the generic coordinate z_q, with $q = 1, \ldots, Q$, is a nonlinear function of the type

$$z_q = \begin{cases} 1, & x_i|x_k - x_h|x_j < 0 \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

for suitable choices of $i, j, k, h \in \{1, \ldots, N\}$. Notice that not every choice of x_i, x_j, x_k, and x_h is meaningful or informative when plugged into expression (9): there are cases where it is invalid (e.g., for $x_i = x_h$ and $x_k = x_j$) and cases where it is redundant (e.g., expressions $x_i|x_k < x_h|x_i$ and $x_i|x_h < x_k|x_i$ are equivalent). The main result of this section is the following proposition, the proof of which is left in the Appendix, which provides the exact definition of the complete nonredundant set of inequalities encoding the sign of all NDI differences from an N-dimensional domain.

Proposition 1: The complete nonredundant set of inequalities encoding the sign of all NDI differences from an N-dimensional domain is given by

$$x_i x_j - x_k x_h < 0 \tag{10}$$

for any $i, j, k, h \in \{1, \ldots, N\}$ fulfilling the constraints $i \leq j$ and $i < k \leq h$, with both $k, h \neq j$. The set consists of exactly

$$Q = \binom{N}{2} + 3\binom{N}{3} + 3\binom{N}{4} \tag{11}$$

inequalities.

Borrowing notation from the theory of quadratic vector equations (QVEs) [48], we have that 2nd-degree monomials can be compactly written as tensor products. As a consequence, transformation φ, for which every coordinate is expressed by an inequality of type (10), can be written in a matrix form as

$$\varphi(\boldsymbol{x}) = \sigma(D(\boldsymbol{x} \otimes \boldsymbol{x})) \tag{12}$$

where $\otimes$ is the Kronecker product, D is a highly sparse finite-difference matrix of size $Q \times N^2$, and σ is the (sign-inverted) vector-valued Heaviside function, i.e., for $\boldsymbol{u} \in \mathbb{R}^Q$

$$\sigma(\boldsymbol{u})_q = \begin{cases} 1, & u_q < 0 \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

for all $q = 1, \ldots, Q$. It is clear that the formulation of expression (12) implies having defined a mapping $q = q(i, j, k, h)$ that relates any choice of $i, j, k, h \in \{1, \ldots, N\}$ fulfilling the requirements expressed in Proposition 1, to every coordinate $q \in \{1, \ldots, Q\}$, in such a way that the expression $u_q = x_i x_j - x_k x_h$ brings no ambiguity. As a matter of fact, it is not essential for the purposes of this section to make a specific choice for this mapping[2], while it is crucial to come up with a convenient choice of it at the implementation level (cfr. with Section III-D). Having that said, matrix D must be compiled at each row $q = q(i, j, k, h)$ by placing entries $1, -1$ at columns $c_1(i, j)$ and $c_2(k, h)$ to encode the coordinate equation $u_q = x_i x_j - x_k x_h$ in such a way that $u_q = (\boldsymbol{x} \otimes \boldsymbol{x})_{c_1} - (\boldsymbol{x} \otimes \boldsymbol{x})_{c_2}$. Considering the order of monomials in $\boldsymbol{x} \otimes \boldsymbol{x}$, we can simply put $c_1(i, j) = c(i, j)$, $c_2(k, h) = c(h, k)$, and

$$D(q(i, j, k, h), c(i, j)) = 1, \quad D(q(i, j, k, h), c(k, h)) = -1 \tag{14}$$

where $c(a, b) := (a - 1)N + b$.

B. Classification Criterion

Before getting into the details, some definitions are in order. First, the set of binary vectors $\{0, 1\}^Q$ must be regarded as $\mathbb{Z}_2$-vector space $\mathbb{Z}_2^Q$, and all binary vectors mentioned hereafter are intended in $\mathbb{Z}_2^Q$. Let us notate $\mathbf{0} = (0, \ldots, 0)^T$ and $\mathbf{1} = (1, \ldots, 1)^T$ the vectors of all zeros and ones, respectively, and $\boldsymbol{e}^j$ be the canonical vectors that are 0 everywhere except for the coordinate j, in which they take value 1, for each $j = 1, \ldots, Q$. Finally, we denote the indexing set of a binary vector $\boldsymbol{a}$, i.e., the set of the indexes of its nonnull values, by $\delta(\boldsymbol{a}) = \{i : \boldsymbol{a}_i = 1\}$.

1) Training: Following the rationale proposed in Section II-C, our aim is to formalize the space that is generated by intersecting all conditions of the type (7) verified over a set of positive samples. For a generic MS value $\boldsymbol{y} \in \mathbb{R}_+^N$, the binary vector $\boldsymbol{w} = \varphi(\boldsymbol{y})$ gathers in its coordinates the sign of every possible mutual NDI difference $w_q = y_i y_j - y_k y_h \lessgtr 0$, taking value 1 or 0 if the corresponding inequality is verified by $<$ or $>$. Let us denote $\boldsymbol{Y} = \{\boldsymbol{y}_m\}_{m=1}^M$ a collection of spectral measurements sampled from the positive class. Following our rationale, coordinate values that are fixed in the set of binary vectors $\boldsymbol{W} = \{\boldsymbol{w}_m\}_{m=1}^M$ are characteristic of the positive class. We then identify the positive class as the set of all binary vectors in $\mathbb{Z}_2^Q$ with

[2]Different choices of the mapping $q = q(i, j, k, h)$ would only result in changing the order of the coordinates in the space $\{0, 1\}^Q$.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

the same fixed coordinate values of $\boldsymbol{W}$. This space is in mathematical terms the affine subspace $\mathcal{A} \subset \mathbb{Z}_2^Q$ generated by $\boldsymbol{W}$, i.e., $\mathcal{A} = \mathcal{A}(\boldsymbol{W}) = \langle \boldsymbol{w}_1, \ldots, \boldsymbol{w}_M \rangle$. To provide an explicit construction of $\mathcal{A}$, if $\boldsymbol{a}^{(b)}$ be the binary vector that stores 1s at every coordinate in which $b = 0$ and 1 is fixed by $\boldsymbol{W}$ and 0s elsewhere, then

$$\boldsymbol{a}^{(1)} = \bigodot_{m=1}^{M} \boldsymbol{w}_m, \quad \boldsymbol{a}^{(0)} = \bigodot_{m=1}^{M} (\boldsymbol{1} - \boldsymbol{w}_m) \tag{15}$$

where $\odot$ is the Hadamard product. If $\boldsymbol{d} = \boldsymbol{1} - \boldsymbol{a}^{(1)} - \boldsymbol{a}^{(0)}$, then $\mathcal{A} = \boldsymbol{a}^{(1)} + \mathcal{V}$, where $\mathcal{V} = \text{span}\{\boldsymbol{e}^i : i \in \delta(\boldsymbol{d})\}$. It follows from this construction that elements of $\mathcal{V}$ are all and only those binary vectors that have 0s at the locations in which coordinate values are fixed by $\boldsymbol{W}$. Translation by $\boldsymbol{a}^{(1)}$ ensures that elements of $\mathcal{A}$ are all and only those binary vectors that share the same fixed coordinate values of $\boldsymbol{W}$.

2) Classification: The construction previously described provides a simple machinery to check whether a binary vector $z \in \mathbb{Z}_2^Q$ belongs to the affine space $\mathcal{A}$ or not, i.e., to classify it. Indeed, one can easily see that $z \in \mathcal{A}(\boldsymbol{W})$ if and only if $z - \boldsymbol{a}^{(1)} = P_{\mathcal{V}}\left(z - \boldsymbol{a}^{(1)}\right)$, where $P_{\mathcal{V}}$ is the canonical projection onto the vector space $\mathcal{V}$, i.e., $P_{\mathcal{V}} = WW^T$, with W being the matrix whose columns are vectors $\boldsymbol{e}^i$, for all $i \in \delta(\boldsymbol{d})$. By linearity of the projection, one has that $z - P_{\mathcal{V}}(z) = \boldsymbol{a}^{(1)} - P_{\mathcal{V}}\left(\boldsymbol{a}^{(1)}\right) = \boldsymbol{0}$, and therefore, classification can be done by checking whether $z = P_{\mathcal{V}}(z)$ or not. Summarizing, given a compound test image $\boldsymbol{X}$, the classification of each pixel $\boldsymbol{x}$ is done by first transforming it into the binary space via $z = \varphi(\boldsymbol{x})$ and then checking its membership to the affine space $\mathcal{A}(\boldsymbol{W})$ by means of the canonical projection. In short, we define a binary output $\boldsymbol{C}$, which provides at each pixel $\boldsymbol{c}$ the binary information

$$\boldsymbol{c} = \begin{cases} 1, & z = \text{WW}^T z \\ 0, & z \neq \text{WW}^T z \end{cases} \tag{16}$$

where $z = \varphi(\boldsymbol{x})$ and output $\boldsymbol{c} = 1$ indicates that pixel $\boldsymbol{x}$ belongs to the target class and the opposite otherwise. Classification can be therefore implemented through basic linear algebra operations mainly involving sparse differences for the transformation φ [see (12)] and matrix–vector multiplications [see (16)]. In terms of computational complexity, the former requires a dedicated discussion, which is given in Section III-D.

C. Learnability

As one can easily see, the proposed one-class model is parameter-free and therefore rigidly and solely dependent on the choice of positive samples. This poses a serious question about how to consider a sample set stable and representative. These are two crucial aspects, which can be related to the concept of learnability of the positive class. The definition of learnability in our model can be expressed by formalizing these two concepts.

1) *Stability:* By including new samples to the training set, there should be a point in which the number of conditions verified throughout the samples stabilizes to a fixed value making irrelevant the inclusion of further samples.
2) *Representativity:* The property of convergence to a fixed value should only depend on the sample set numerosity and not on the specific choice of the samples.

Regarding stability, if we notate $\psi^{(b)}$, for $b = 0$ and 1, the number of coordinates in which the value b is fixed by $\boldsymbol{W}$, then we easily have $\psi^{(b)} = \sum_{i=1}^{Q} \boldsymbol{a}_i^{(b)}$ and the number of conditions verified throughout the samples is then $\psi = \psi^{(0)} + \psi^{(1)}$. In geometrical terms, ψ is the codimension of the affine space $\mathcal{A}$ in $\mathbb{Z}_2^Q$, and therefore, the proposed stability condition is a requirement that this value can be determined robustly with a limited amount of samples. We can observe that, by construction, the number ψ is a (nonstrictly) decreasing function[3] of the number of samples M and therefore theoretically convergent (possibly to zero) for M that tends to infinity. What makes the target class learnable is then the verification of the convergence of the function $\psi(M)$ to some fixed positive value given a reasonably limited sample set numerosity, say M'.

To guarantee representativity, one has to ensure that this convergence property is general and it is verified for different choices of sample sets of the same numerosity M'. This can be done by choosing, for every M, a predetermined number R of random subsets of numerosity M and checking whether the average value $\bar{\psi}(M) = \sum_{r=1}^{R} \psi_r(M)/R$ reaches convergence in correspondence of some value M'. Controlling convergence in terms of relative variation, we can then formalize learnability as follows.

Definition 1 (Learnability): The positive class is considered learnable if there exists a sample numerosity M' such that the average relative variation of $\bar{\psi}(M)$ approaches zero within a fixed tolerance, i.e.,

$$\left\| \frac{\bar{\psi}(M) - \bar{\psi}(M-1)}{\bar{\psi}(M)} \right\| < \text{tol} \tag{17}$$

for all $M > M'$.

D. Implementation Remarks

It is worth emphasizing here the size Q of the binary space in which spectral vectors $\boldsymbol{x} \in \mathbb{R}_+^N$ are mapped by the transformation φ, which is hyperdimensional if compared to the number N of spectral bands available for classification. To better illustrate this, in Table I, some values returned by expression (11) are listed. A practical consequence of these sizes is that the implementation of the proposed one-class classification model requires specific handling of computing resources such as RAM and parallel use of CPUs. Indeed, although the classification algorithm is relatively simple in terms of complexity of the operations involved (only multiplications, differences, and sign checking), there is a significant computational burden associated with the number of conditions that need to be verified, which is, of course, bounded from above by Q. However, the training and classification tasks scale up very differently. More specifically, training

[3]It counts the number of stable values (obtained by intersection) across a set of binary vectors, which increases in size.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

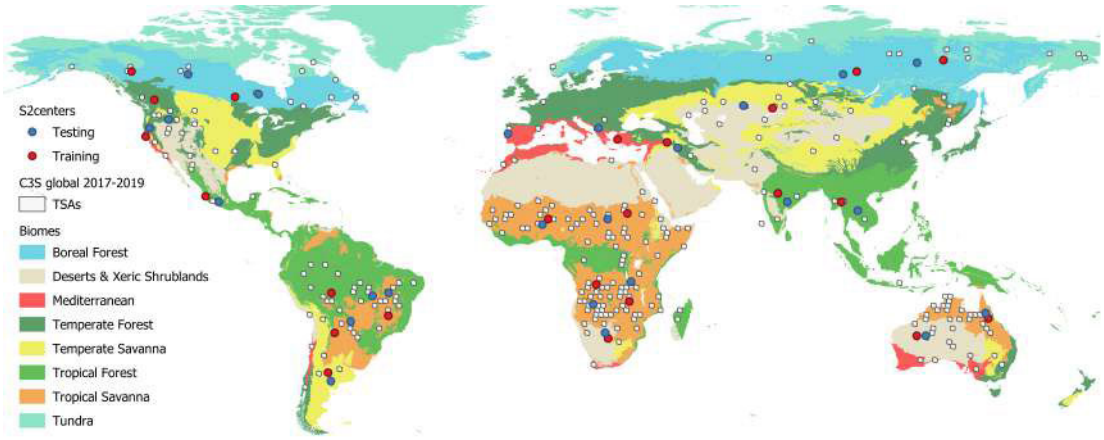


Fig. 1. Global overview of the validation experiment that illustrates the spatial distribution of C3S TSAs, the biomes classification, and the final choice of training/testing sites.

TABLE I
EXAMPLES OF BINARY SPACE DIMENSION AS A FUNCTION OF THE NUMBER OF SPECTRAL BANDS

N	$Q(N)$
4	21
10	1035
20	18145
50	750925
100	12253725

requires checking all Q relations but only at each sampled pixel spectral value $\boldsymbol{y}_m$, for $m = 1, \ldots, M$, and this does not present a relevant issue for implementation.

On the other hand, classification requires checking only relations that are fixed by the sample set $\boldsymbol{W}$, but on all image pixels $\boldsymbol{x}$ from image $\boldsymbol{X}$. Recalling that the number of fixed conditions is ψ, as defined in Section III-C, we see by the argument above that this number is critical from the implementation viewpoint in that many bandwise comparisons are required for the classification step. To better illustrate this, the pseudocode of Algorithm 1 in the Appendix describes how transformation φ can be implemented on a machine, for the classification case. Notice that in the pseudocode, the expression $z_q = (a < b)$ is a shorthand for $z_q = 1$ if $a < b$ or $z_q = 0$ otherwise. We can see that bandwise comparisons are controlled by the variable $\boldsymbol{d}$, as defined in Section III-B1, which takes the value 0 at the coordinates q that are fixed by the sample set $\boldsymbol{W}$. A concluding remark is needed concerning the way in which nested `for` loops are organized in Algorithm 1. There are several ways in which the four indices i, j, k, and h can be scanned, and the proposed one allows to store a temporary variable (x_{temp}) in the RAM, which is fixed in many subsequent iterations, therefore allowing to save a considerable amount of computational time.

IV. EXPERIMENTAL VALIDATION

Producing reference information for validation of RS global products requires huge efforts and literature lacked of robust and globally assessed datasets for many years. The BA detection case is by far one of the most addressed in the literature and currently the only type of vegetation change for which well-assessed globally scaled validation datasets for MS algorithms have been made available. The purpose of this section is to assess the proposed one-class classification algorithm on the challenging RS problem of BA detection at the global scale using satellite-borne MS S2 imagery. Only in 2020, the Burned Area Reference Database (BARD), the first publicly available database created by compiling existing reference BA datasets from different international projects, was created as one of the outcomes of the ESA FireCCI project [49]. However, global datasets in BARD were formerly available for years in which S2 was not operative yet (i.e., prior to 2016) and it is only very recent the publication and inclusion in BARD of the first S2 temporally compatible global dataset covering years 2017–2019. This dataset is called C3S_global_2017-2019[4] [50], for brevity C3S in the sequel, and it consists of long temporal reference shapefiles analyzed using L8 image pairs over 316 Thiessen scene areas (TSAs) sampled according to a regrouped 8-biome stratification [51]. The C3S is used in [50] to assess the spatial accuracy of three major global-scale BA products, namely, FireCCI51, C3SBA10, and MCD64C6, using long temporal reference units to diminish the influence of temporal reporting (i.e., dating errors) on the spatial accuracy estimates.

In this section, we are going to apply both the proposed and the OC-SVM methods in a subset of locations from C3S data and compare the accuracy obtained. To maximize the compatibility between the three global-scale BA products, our results, and the temporal strategy adopted in [50] for validation (details are recalled in Section IV-B1), we applied our bitemporal method in a nearly operative multitemporal fashion to several pairs of images with final aggregation of the results (the temporal strategy is detailed in Section IV-B3). A significant part of the validation work presented hereafter is concerned with the adaptation of C3S reference information to S2 data and the generation of a comprehensive training/testing dataset of S2 images. This activity is described in Section IV-B. The outcome of the experiments and the discussion of the results are finally given in Section IV-D. An overview of the spatial distribution of C3S TSAs, the 8-biome classification, and the training/testing locations of our validation experiment is provided in Fig. 1. Throughout this section, MS images are displayed in false-color composition where green and blue

[4]Downloadable at: https://doi.org/10.21950/BBQQU7

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

TABLE II
S2 SPECTRAL BAND PROPERTIES

Name	Wavelength	Description	Spatial res.	index (b)
B2	493nm	Blue	10m	1
B3	560nm	Green	10m	2
B4	665nm	Red	10m	3
B5	704nm	EDGE 1	20m	5
B6	740nm	EDGE 2	20m	6
B7	783nm	EDGE 3	20m	7
B8	833nm	NIR	10m	4
B8A	865nm	EDGE 4	20m	8
B11	1610nm	SWIR 1	20m	9
B12	2190nm	SWIR 2	20m	10

channels of RGB are replaced by NIR and SWIR to render green vegetation and facilitate visual discrimination of BAs (resulting in shades of purple color).

A. S2 Data

S2 MS imagery is chosen for assessment of BA detection one-class classification algorithms because of its spectral content similar to Landsat-8 data, but with more vegetation-related spectral bands available, improved spatial resolution (L8 products are at 30-m resolution, while S2 useful bands for vegetation change studies are at 10-/20-m resolution) and finer temporal resolution (L8 repeats every 16 days, while S2 in between two and five days). Salient properties of S2 spectral bands are summarized in Table II. Notice that 60-m resolution bands are discarded from our model being them mostly appropriate for cloud screening and atmospheric correction and not useful for vegetation-related studies. To fit all ten selected spectral bands into a common spatial grid, 10-m resolution bands are degraded to 20-m resolution by local averaging. The spatial size of each band is therefore 5490 × 5490 pixels. Concatenation of two images produces a compound image $\boldsymbol{X}$ with $N = 2B = 20$ spectral bands, which corresponds to a dimension of the binary space $Q = 18\,145$, according to Table I.

B. Experimental Setup

To address the technical details related to testing the proposed one-class classification algorithm for BA detection, the experimental part of this work is articulated into three main activities, which are summarized here and further developed in the remainder of this section.

1) *Test Dataset:*
   a) Select a spatial subset of the BARD reference database that is globally representative of its different biomes.
   b) Build a test dataset of S2 time series that intersect with selected validation locations.
   c) Adapt C3S L8-based reference information to make it compatible with S2 test data.
2) *Training Dataset:*
   a) Build a training dataset of image pairs representative of the different biomes considered in BARD.
   b) Define a sampling strategy and extract a stable set of training samples.

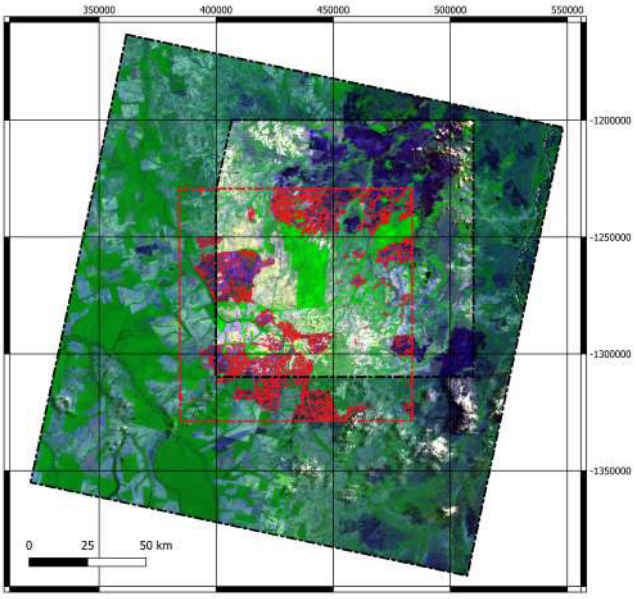


Fig. 2. Test site example. The bigger image is a reference Landsat-8 image, the smaller one is a selected S2 image, and both are contoured in the black dotted-dashed line. In the red solid line, the C3S shapefile with its contour in the dotted-dashed line. The final test site extension is found by intersecting the S2 image footprint and the C3S product contour.

3) *Validation Strategy:*
   a) Establish a multitemporal scheme to apply bitemporal BA classification to the S2 time series.
   b) Define appropriate validation metrics and compare with other global-scale products.

1) Test Dataset: To better illustrate the process that drives the generation of testing/training datasets, it is important to recall some properties of C3S data. In C3S [50], reference shapefiles are generated by a visual analysis of every L8 image available within temporal intervals ranging between 48 and 368 days. Spatial covering of the reference products is restricted to frames of size 100 × 100 km centered with respect to the corresponding TSAs (see Fig. 2). Time series of images are first analyzed by visual interpretation (see details in [50]) on consecutive temporal pairs and three classes are identified: 1 = burned, 2 = no-data, and 3 = unburned, with no-data class including clouds and shadows. Then, burned and no-data pixels from all analyzed pairs are accumulated and finally converted to a polygon and included in the reference shapefiles. It is important to emphasize here that, as a consequence of the abovementioned accumulation criterion, which is explained and visualized in [50, Fig. 6]), some BAs are potentially missed in the final reference products because of the accumulation of no-data pixels that are mapped in the same location of burned ones, but after the burn event.

To perform the experiments, we selected a subset of 24 reference locations that is representative of the spatial distribution and biome stratification of all the sites in C3S. The number of testing sites over each biome is thus approximately proportional to the world fraction of the burnable surface of that biome. The final outcome of this choice is shown in Fig. 1 and detailed in Table III. C3S reference products are not designed

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

TABLE III
DATASET OF TESTING SITES AND ASSOCIATED TIME SERIES OF S2 IMAGES

Tile	Country	Biome	Time span	images
T10TEM	USA (CA)	Temperate Forest	27/06/2018 - 11/08/2018	3
T11TNK	USA (ID)	Temperate Forest	11/07/2018 - 04/09/2018	3
T12VWL	CAN (AB)	Boreal Forest	17/05/2019 - 09/09/2019	3
T14QLF	MEX	Tropical Forest	12/01/2019 - 12/05/2019	6
T16UDD	CAN (ON)	Boreal Forest	11/06/2018 - 20/08/2018	2
T20HLC	ARG	Temperate Savanna	11/12/2016 - 01/03/2017	4
T21KUU	PRY	Tropical Savanna	16/06/2017 - 19/09/2017	4
T22LDN	BRA (MT)	Tropical Forest	31/05/2019 - 23/09/2019	6
T23LLJ	BRA (TO)	Tropical Savanna	29/06/2019 - 17/09/2019	5
T29TNE	PRT	Mediterranean	04/06/2017 - 18/08/2017	4
T31PDN	BEN	Tropical Savanna	24/01/2019 - 30/03/2019	3
T34KFV	BWA	Deserts, Xeric Shrub	30/07/2018 - 22/11/2018	5
T34LBK	AGO	Tropical Savanna	27/05/2019 - 19/10/2019	7
T34PGV	SDN	Tropical Savanna	11/12/2016 - 31/03/2017	4
T34TDM	ALB	Mediterranean	01/06/2017 - 20/08/2017	4
T36MTT	TZA	Tropical Savanna	09/06/2018 - 02/10/2018	7
T38SNE	IRQ	Temperate Forest	08/06/2018 - 26/09/2018	6
T41UQQ	KAZ	Deserts, Xeric Shrub	01/05/2017 - 09/08/2017	3
T44QMF	IND	Tropical Forest	03/02/2019 - 24/05/2019	6
T47VMF	RUS (KYA)	Boreal Forest	07/06/2019 - 10/09/2019	4
T48PTC	THA	Tropical Forest	04/01/2018 - 10/03/2018	3
T51KXP	AUS (WA)	Deserts, Xeric Shrub	06/08/2018 - 24/11/2018	5
T51VVK	RUS (SA)	Boreal Forest	31/07/2017 - 14/09/2017	3
T54KYG	AUS (QL)	Tropical Savanna	01/07/2019 - 14/10/2019	4

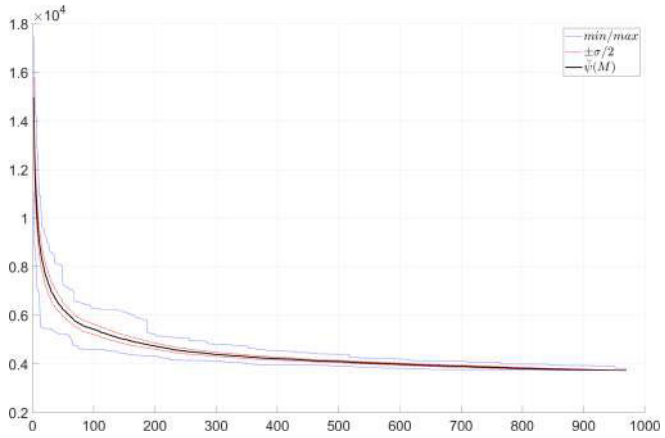


Fig. 3. Learning curve of the BA class. The graph shows the convergence of the $\bar{\psi}(M)$ function (black solid line), which shows relative variation below tol $= 10^{-6}$ at sample numerosity $M' = 970$. The average $\bar{\psi}$ is computed out of $R = 50$ randomly extracted subsets from the initial pool of $M'' = 2000$ samples. The statistical meaning of the average is enforced by showing in the plot min/max extremes (dashed blue lines) and the σ-wide symmetric interval around the mean (red dotted-dashed line).

for S2 data, the most critical discrepancies being spatial and temporal allineations. To build a test dataset of S2 time series, we therefore made some choices regarding these two aspects. From the spatial viewpoint, since S2 images are organized into MGRS 100 × 100 km tiles, we selected the S2 tiles that overlapped most with the corresponding C3S products. This is illustrated through an example in Fig. 2. Regarding the temporal aspect, two choices are made. First, given the diverse temporal range of the products in C3S, we kept as a reference a time span of approximately four months. Second, for each test site, we built short time series of images keeping the first/last S2 images temporally as close as possible to the first/last ones in the corresponding L8-based reference time series and including intermediate images so that each series consists of approximately ∼1.5 image/month (i.e., ∼1 image every 20 days). The numbers of images below this threshold are due to heavy cloud coverage that made it impossible to find clear images. The average time interval in the time series is 21.9 days. Spatial and temporal details of the test dataset are summarized in Table III. In order to make the reference information compatible with S2-derived BA products, for each validation site, the portion of C3S shapefile that intersected the S2 tile footprint was converted into the corresponding UTM projection[5] and then rasterized to the same 20-m resolution grid of BA products to allow for pixelwise comparison.

2) Training Dataset: In producing a training dataset for the experiments, two main aspects have been considered: 1) how images are selected and 2) how images are sampled. The image selection process involves considering both spatial and temporal characteristics of the images in the time series used for testing (cfr. with Section IV-B1). Spatially, for each testing site, a representative training site is selected, which is included in the same continental extension of the underlying biome. Training and testing sites are shown in Fig. 1 for a visual assessment of their geographical locations. Temporally, images selected for the training are chosen in the same temporal period of C3S products, i.e., between years 2017 and 2019, and a different year is always chosen with respect to the one selected for the test case. For each training site, an image pair is formed by selecting two images (preevent and postevent) that are approximately 30 days apart (this choice is a consequence of the multitemporal scheme described in Section IV-B3), with a few exceptions due to heavy cloud coverage in some training sites. The collection of 24 training image pairs is then sampled randomly and uniformly in space within the boundaries of the BA reference shapefiles to produce a dataset of training observations of the BA class. Stability, representativity, and numerosity of the training set are determined following the concept of learnability described in Section III-C. Starting with a set of randomly extracted 2000 pixel locations across all the 24 training sites (∼83 samples for each training site), we estimated the convergence of the $\bar{\psi}(M)$ function (cfr. Definition 1) within a tolerance tol $= 10^{-6}$ for a sample numerosity of $M' = 970$. The learnability of the BA class is shown in Fig. 3 by showing the convergence of the $\bar{\psi}(M)$ function. To perform the training of the classifier, i.e., the definition of the affine space $\mathcal{A}(\varphi(\boldsymbol{Y}))$, a final training set $\boldsymbol{Y}$ of numerosity $M' = 970$ is extracted randomly from the initial pool of samples. With these samples, the estimated codimension of the affine space representing the BA class is $\psi = 3720$.

3) Validation Strategy: As described in Section IV-B1, time series in the test dataset consist of sequences of images that are sampled approximately every 20 days. Within the considered time frame of each test site (∼4 months), BAs are identifiable at different times according to the temporal progression of fires. In order to maximize the probability of detecting all of them, a multitemporal scheme that extends the bitemporal detection approach to several image pairs is required. Let us denote by $\boldsymbol{X}^1, \ldots, \boldsymbol{X}^T$ a sequence of images, with $T \geq 2$ the number of images in it. In general, for any acquisition time t, with $1 \leq t < T$, the corresponding image can be paired

[5]C3S shapefiles are in sinusoidal projection.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CONFUSION MATRIX FOR THE BINARY CLASSIFICATION BA DETECTION PROBLEM

		Reference	
		burned	unburned
Product	burned	A_{11} (TP)	A_{12} (FP)
	unburned	A_{21} (FN)	A_{22} (TN)

with all its successive acquisitions $t+1,\ldots,T$. However, as described in the previous section, the representativity of training samples for this experiment is bounded to time intervals that are approximately 30 days. Therefore, for each test site, we apply a multitemporal scheme that involves pairing each image $\boldsymbol{X}^t$ only with its two closest successors $\boldsymbol{X}^{t+1}$ and $\boldsymbol{X}^{t+2}$, thus involving temporal intervals of 20 and 40 days. For each test site, all the BA maps produced with this multitemporal scheme (with T images all possible pairs involving at most the two closest successors are exactly $2T-3$) are finally merged together and their aggregation is the final BA map $\boldsymbol{C}$ that is confronted with the corresponding reference map in order to evaluate the accuracy of the detection.

To evaluate the performance in terms of spatial accuracy, the metrics proposed in [50] are used, which are in line with the guidelines of GCOS and very effective for the evaluation of binary maps with a minority class. These metrics are commission error (CE), omission error (OE), relative bias (RB), and the $F1$-score (also known as dice coefficient). All these metrics are obtained as functions of the confusion matrix entries, also known as true positives (TPs), false positives (FPs), false negatives (FNs), and true negatives (TNs), see Table IV. Explicitly, metrics formulas are:

1) $\mathrm{CE} = (\mathrm{FP}/\mathrm{TP}+\mathrm{FP})\cdot 100$.
2) $\mathrm{OE} = (\mathrm{FN}/\mathrm{TP}+\mathrm{FN})\cdot 100$.
3) $\mathrm{RB} = (\mathrm{FP}-\mathrm{FN}/\mathrm{TP}+\mathrm{FN})\cdot 100$.
4) $F1 = (2\mathrm{TP}/2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})\cdot 100$.

These values are given as percentages. To compare the BA detection results of OC-SVM and the proposed method with state-of-the-art global-scale BA products, we take advantage of the assessment provided in [50] for the three major global BA products FireCCI51, C3SBA10, and MCD64C6 against the same C3S reference data. These products are described in detail in [50] and references therein.

C. Training of the One-Class SVM

An efficient implementation of OC-SVM is `OneClassSVM` from the `scikit-learn` Python library. The OC-SVM model requires some choices to be made a priori about its kernel and parameters [26]. We select the radial basis function (RBF) kernel with the default parameter γ set by the algorithm, which corresponds to the inverse between the product of the number of features (20) and the variance of the input data. For the choice of the ν parameter, which sets an upper bound on the fraction of training errors and a lower bound of the fraction of support vectors [52], several models are trained on the training data obtained in Section IV-B2 for a range of values $\nu \in (0,1)$ and the best overall performer, based on the $F1$-score, was found for $\nu = 0.1$ [see Table VI]. More details are given in Section IV-D.

TABLE V
GLOBAL ACCURACY ESTIMATES (%) OBTAINED BY THE BEST PERFORMING OC-SVM AND THE PROPOSED METHOD, COMPARED WITH THE THREE GLOBAL-SCALE BA PRODUCTS ASSESSED AGAINST THE SAME REFERENCE DATA IN [50]

		F1	CE	OE	RB
Product	C3SBA10	62.9	17.2	49.3	-38.7
	FireCCI51	66.7	19.4	43.1	-29.4
	MCD64C6	62.5	18.7	49.3	-37.6
Method	OC-SVM	69.6	35.7	**24.1**	17.9
	Proposed	**73.7**	**12.3**	36.4	-27.5

TABLE VI
CHOICE OF THE BEST $F1$-SCORE (%) PERFORMING OC-SVM MODEL TRAINED ON THE 970 POSITIVE SAMPLES FOR DIFFERENT ν VALUES. THE NUMBER OF CORRESPONDING OUTLIERS RETURNED BY EACH MODEL IN CLASSIFYING TRAINING DATA IS ALSO REPORTED

ν	0.1	0.01	0.001	0.0001
F1	**69.6**	54.6	53.6	53.9
outliers	97	13	7	13

D. Results and Discussion

Each global-scale simulation performed in the experiments is run over a set of 104 images (cfr. with Table III). Each image has $5490 \times 5490 \times 10$ pixels and data are stored as 16-bit unsigned integers; therefore, the total amount of data processed is 58.24 GB. Data processing is performed in Python on a home workstation with hardware Intel[6] Xeon[6] CPU E3-1240 v2 3.40GHz, 4C/8T, 32-GB RAM. According to the multitemporal strategy described in Section IV-B3, for each test site, several image pairs are processed and their results are aggregated to get the final BA classification maps. With a grand total of 136 image pairs processed, each one taking ~2.5 min to complete, the entire dataset is processed in nearly 6 h.

The main figure for assessment of the BA detection exercise is given in Table V, which presents the accuracy metrics obtained by the best performing OC-SVM and the proposed method alongside with those reported for C3SBA10, FireCCI51, and MCD64C6 in [50]. During the training stage of OC-SVM method, the best performer was obtained for $\nu = 0.1$, which corresponds to an $F1$-score of 69.6%, as shown in Table VI. It is interesting to note that this model is highly conservative as it classifies 97 out of 970 training samples (exactly one-tenth) as outliers, i.e., unburned. We can see from Table V that the proposed method outperforms the selected OC-SVM and the global products significantly, by showing the higher $F1$-score (73.7%) and the lowest CE (12.3%). The lowest OE is obtained by OC-SVM (24.1%), which is, however, at the cost of the highest CE that amounts to 35.7%, almost three times the CE of the proposed method and two times the CE of the global products. Accuracy estimates obtained by the best performing OC-SVM and the proposed method can be compared at a finer level over each test site by inspection of Table VII. Spatial location of correct and

[6]Registered trademark.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

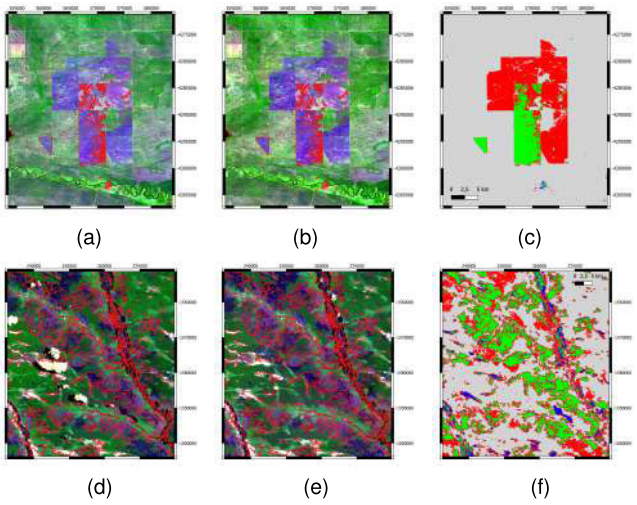


Fig. 4. Two details of overestimated CE reported in our experiments due to BA spots not represented in the reference data used for validation: (a)–(c) site T20HLC and (d)–(f) site T34LBK. For validation maps [see (c) and (f)], the legend is the same as in Figs. 7 and 8. In both cases, reported false alarms (red pixels) are actually correct detections. (a) L8 (03/03/2017). (b) S2 (01/03/2017). (c) Validation map. (d) L8 (03/03/2017). (e) S2 (01/03/2017). (f) Validation map.

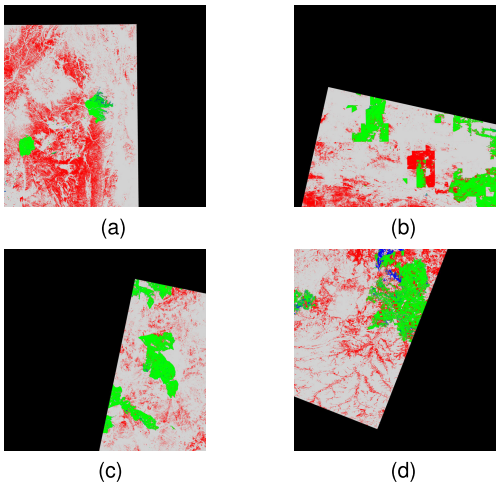


Fig. 5. Critical false alarms reported for the OC-SVM method. Validation maps at four test sites: (a) T11TNK, (b) T20HLC, (c) T41UQQ, and (d) T51VVK. For validation maps, the legend is the same as in Figs. 7 and 8.

erroneous detections of the proposed method can be visualized in Figs. 7 and 8, with the validation maps of each test site.

With respect to the real situation on the ground, the actual CE obtained by the two tested methods is slightly better than the reported value. Indeed, as remarked in Section IV-B1, because of the aggregation technique used in [50] to generate the C3S reference, a potential was reported in missing some BA spots if occlusions, such as clouds and shadows, were accumulated in temporal instants after the burn event. After a careful check of the final BA maps, we realized that this happened in many spots, and the proposed method revealed many of those. More specifically, the majority of false detections of the proposed method, the red pixels in the validation maps shown in Figs. 7 and 8, are actually true BAs and thus correct detections. Two clear examples of this situation are

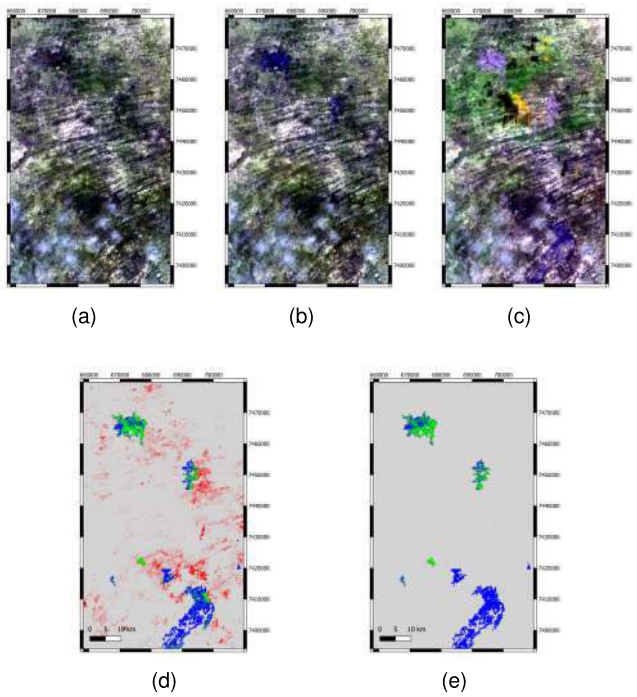


Fig. 6. Particular of missed detections in the T34KFV site (Botswana). The sequence shows: (a)–(c) three consecutive portions of S2 images where the BAs first appear (13/09/2018, 03/10/2018, and 22/11/2018, respectively), and validation maps for (d) OC-SVM and (e) proposed method. For validation maps [see (d) and (e)], the legend is the same as in Figs. 7 and 8.

shown in the particulars shown in Fig. 4, where for two spots in the test sites T20HLC and T34LBK, we both show: the reference L8 image used to generate the reference shapefile, the temporally closest image in the test time series of S2 images, and the validation map produced by our method. We can clearly see from Table VII that the reported CE (23.3% and 30.1% for T20HLC and T34LBK, respectively) should be corrected toward zero. This is not the case for OC-SVM, for which the high CE remains critical, as clearly visible in Fig. 5.

The proposed method outcome confirms the trend of many BA products that missed burned spots are more critical than false detections being RB mostly negative, thus having OE significantly greater than CE. Notice that OC-SVM behaves exactly in the opposite way, with bias toward false detections. From Table VII, we see that OC-SVM shows lower OE than the proposed method in almost every test site, however, as already mentioned, at the cost of critically high false detection rates almost everywhere. Considering an omission rate of ∼45% as a common reference (the reported OE for the other three BA global products is in this range), we see from Table VII that our method performed better in 18 out 24 test locations, and returned critical values OE $>$ 70% in three test sites: T34KFV and T51KXP, corresponding to Deserts/Xeric Sh. biome in Africa and Australia, respectively, and T47VMF covering the Boreal Forest biome, in Russia. This can be visually assessed looking at the validation maps in Figs. 7 and 8, with blue pixels indicating missed detections. Extremely dry areas, such as those in the Deserts/Xeric Sh. biome, typically put BA detection algorithms at the edge, mostly because vegetation is dry and weakly green before the fire [53], thus

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

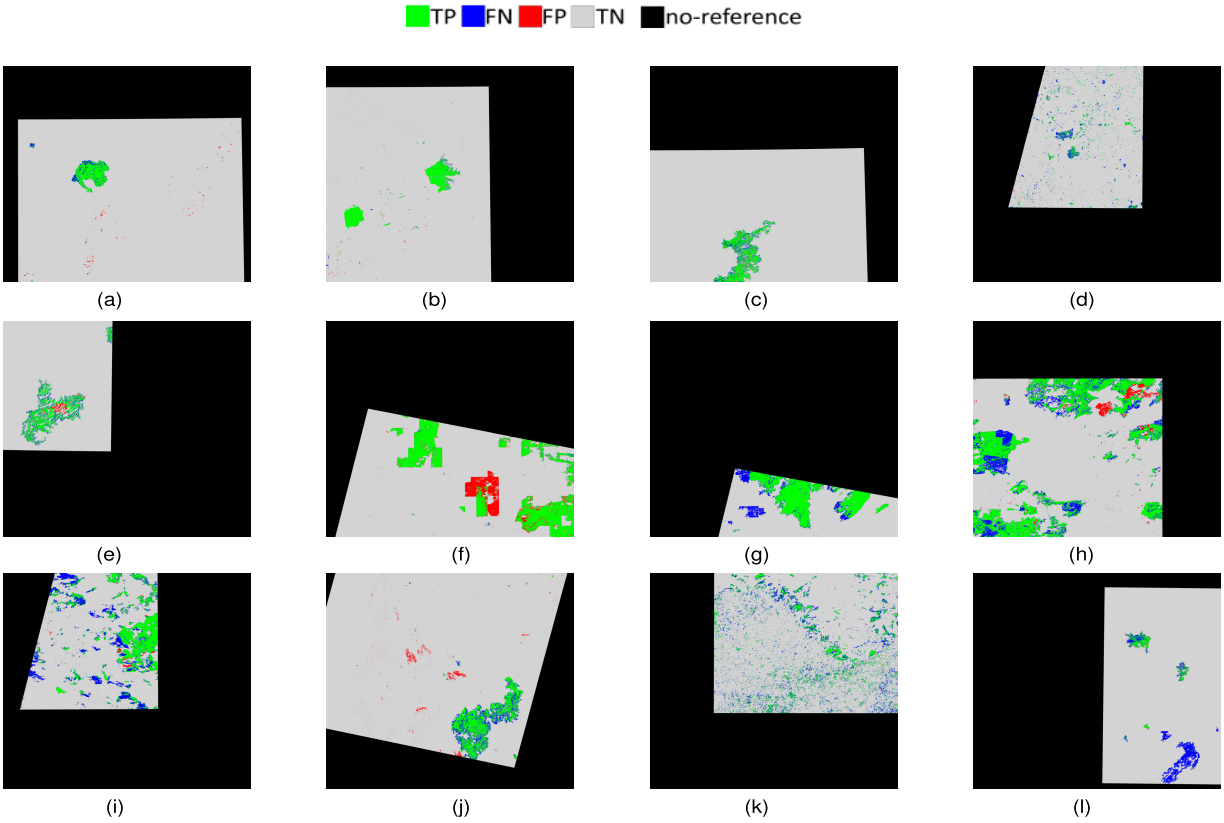


Fig. 7. BA detection validation maps illustrating correct and erroneous detections at the pixel level for test sites 1 to 12 (cfr. with Table III) for the proposed method. (a) T10TEM. (b) T11TNK. (c) T12VWL. (d) T14QLF. (e) T16UDD. (f) T20HLC. (g) T21KUU. (h) T22LDN. (i) T23LLJ. (j) T29TNE. (k) T31PDN. (l) T34KFV.

TABLE VII

ACCURACY ESTIMATES (%) OBTAINED BY THE BEST PERFORMING OC-SVM AND THE PROPOSED METHOD, DETAILED AT EACH TEST SITE

	OC-SVM				Proposed			
Tile	F1	CE	OE	RB	F1	CE	OE	RB
T10TEM	50.6	64.5	12.1	147.9	85.3	9.8	19.1	-10.4
T11TNK	16.6	90.8	8.0	904.7	93.1	6.3	7.5	-1.3
T12VWL	68.7	40.1	19.4	34.5	85.1	4.4	23.3	-19.8
T14QLF	68.3	29.3	34.0	-6.7	64.1	8.7	50.6	-45.9
T16UDD	87.4	9.2	15.7	-7.2	80.7	8.1	28.0	-21.7
T20HLC	66.0	50.4	1.3	98.9	85.7	23.3	2.8	26.7
T21KUU	86.0	13.7	14.2	-0.6	87.6	0.2	21.9	-21.7
T22LDN	80.8	10.2	26.5	-18.2	80.4	8.3	28.3	-21.8
T23LLJ	75.8	6.5	36.2	-31.8	74.6	4.3	38.9	-36.2
T29TNE	64.6	36.7	33.9	4.4	76.6	14.3	30.7	-19.1
T31PDN	72.7	13.0	37.6	-28.3	66.5	5.0	48.8	-46.1
T34KFV	28.2	75.7	66.2	39.3	41.2	7.6	73.5	-71.3
T34LBK	72.1	33.3	21.5	17.7	70.5	30.1	28.8	1.9
T34PGV	83.6	20.1	12.3	9.7	86.0	13.0	15.1	-2.4
T34TDM	40.4	65.4	51.4	40.7	59.6	39.9	40.9	-1.7
T36MTT	83.9	19.2	12.8	7.9	81.0	9.3	26.8	-19.2
T38SNE	38.8	74.9	14.7	239.9	76.0	11.6	33.4	-24.7
T41UQQ	74.5	39.6	2.8	60.9	95.9	1.2	6.9	-5.8
T44QMF	77.0	19.9	25.9	-7.5	57.9	13.8	56.5	-49.5
T47VMF	59.6	6.4	56.3	-53.4	38.5	0.5	76.1	-76.0
T48PTC	66.6	30.6	36.0	-7.8	66.7	21.6	41.9	-25.9
T51KXP	81.7	4.9	28.4	-24.7	30.2	1.7	82.2	-81.8
T51VVK	58.3	56.4	12.2	101.3	75.3	23.9	25.4	-1.9
T54KYG	80.1	0.2	33.0	-32.9	79.4	0.0	34.1	-34.0

producing unnoticeable differences in some of the spectral bands correlated with water content and chlorophyll activity. However, the detection maps at site T34KFV show that missed detections are coherent in space being the missed spots mainly the BAs in the bottom part of the tile, as shown in detail in Fig. 6, for both tested methods. This and the fact that in the test site T41UQQ, which belongs to the Deserts/Xeric Sh. biome in Kazakhstan, the performance is nearly optimal suggesting that the problem of missed detections may depend on an underrepresentation of the BA class in the training data for this biome. The same appears to be true for the case of the high OE related to the Russian site T47VMF, in the Boreal Forest, which is the only critical one out of four cases in the same biome with the other three performing nearly optimally as well. To conclude our quantitative discussion of the results, we want to stress here the fact that the proposed validation exercise has not been designed to maximize the performance of the algorithm nor to test it operationally, a task that would require a wider reference dataset and a comprehensive training campaign driven by the support of climate and forest scientists. Given the novelty of the proposed approach, this experimentation shows instead that, even in a pretty much straightforward application scenario (the proposed training phase can be performed by anyone, even nonexperts in the field), the results that can be obtained are substantially of higher quality than those reported for reference products.

There are also other aspects emphasizing the novelty and effectiveness of the proposed model, both of methodological and technical character. First, the training required a sample size of only 970, and this is a remarkably small size, given that experimentation covers several locations around the globe

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

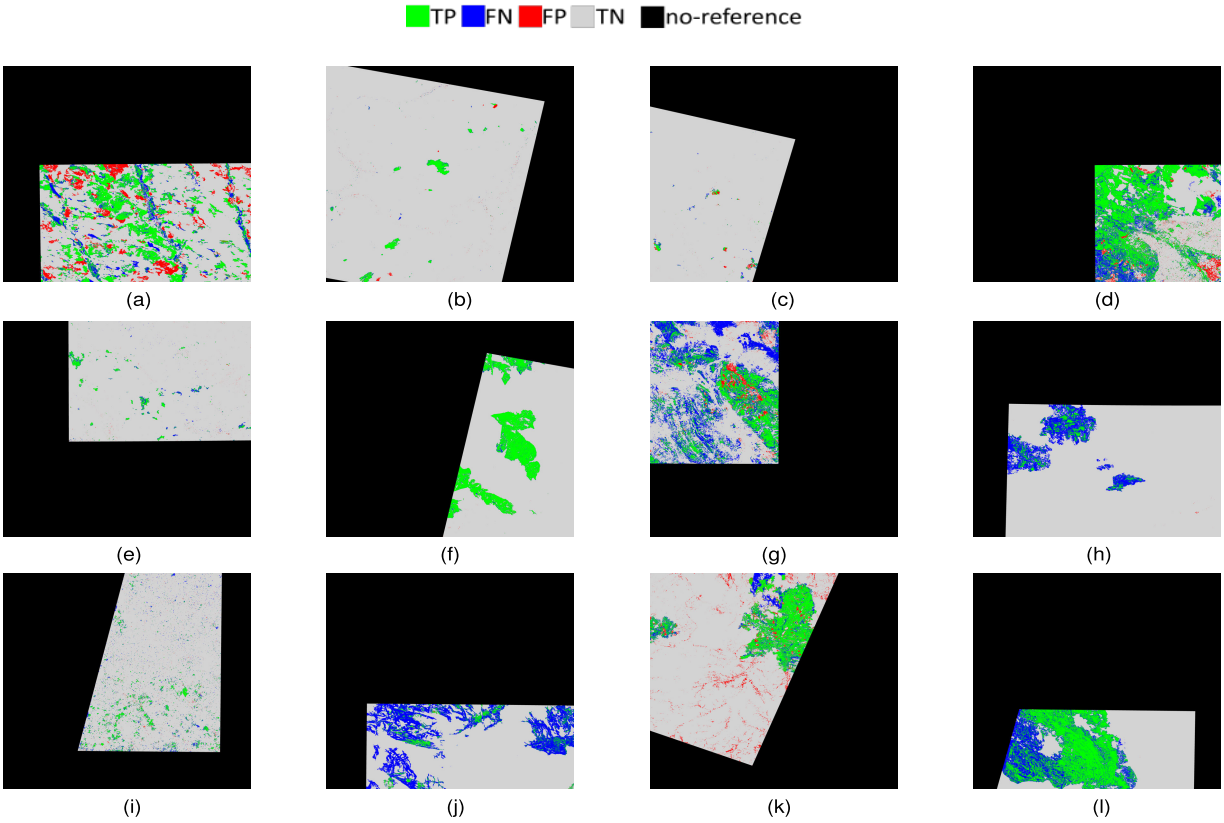


Fig. 8. BA detection validation maps illustrating correct and erroneous detections at the pixel level for test sites 13 to 24 (cfr. with Table III) for the proposed method. (a) T34LBK. (b) T34PGV. (c) T34TDM. (d) T36MTT. (e) T38SNE. (f) T41UQQ. (g) T44QMF. (h) T47VMF. (i) T48PTC. (j) T51KXP. (k) T51VVK. (l) T54KYG.

with data at 20-m spatial resolution. Moreover, samples are only needed for the positive class, making their collection a relatively simple task. Another key feature is that, unlike many other BA detection methods, the proposed approach does not rely on auxiliary thematic products of any sort such as cloud maps, water bodies, and nonburnable objects in general, neither of ancillary information like AFs to train locally driven parameter estimation models [50]. The proposed model is applied globally, with no need of local modeling of burned vegetation characteristics and/or seasonal trends. A final remark is regarding the temporal aspect. In order to focus our sample set representative capability to a specific temporal target, in our test dataset, we reduced the temporal resolution of the time series to nearly 20 days, a considerable degradation with respect to the nominal resolution of 2–5 days of S2. On the one side, the downside of MODIS-based BA products, such as C3SBA10, FireCCI51, and MCD64C6, is the spatial resolution ranging from 250 to 1000 m, which is, however, balanced by the fine temporal resolution, i.e., two days, and high spectral detail with 20 reflective and 16 thermal bands. Considering the temporal attenuation of burn effects on vegetation, we see that state-of-the-art products based on MODIS should take advantage of higher probabilities of finding nonoccluded pixels even in heavily cloudy locations. On the other side, the fact that even in the proposed limited setup, the proposed model performs significantly better than the competitors, in general but especially for the omission rates, indicates that the BA maps produced by our method can potentially reveal many burned spots that are typically missed using other procedures. To conclude, we recall that training sample collection has been restricted to the minimum effort (one training site for one test site) with the main consequence of reducing the probability of sampling the whole range of burned vegetation types. In a broader, more operationally oriented, validation exercise, all these aspects can be addressed with less restrictions possibly involving climate and forest scientists to aid the training phase and leading to further improvements.

V. Conclusion

One-class classification can be extremely useful in application contexts where the class of interest is minority compared to the background. For MS imagery, which is typically characterized by a large number of spectral bands, class representation often requires abstract domain transformation techniques to enhance discrimination. However, to catch the intrinsic complexity of certain classes, some limitations are often encountered: 1) known one-class classification models are not sophisticated enough to handle cases of very pronounced intraclass variability and 2) more powerful NN models, which can fit a very broad variety of problems, require many parameters to be determined a priori and significantly large sets of training samples. In this article, a novel one-class

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

classification model has been proposed, which specifically addresses these limitations. The model retains a high level of abstraction as it embeds the MS domain into an abstract hyperdimensional binary space, which is built mathematically by generalizing an empirically observed principle. The proposed model makes particular sense for MS image classification as it exploits certain interesting mathematical properties of normalized differences, a family of spectral indexes that in many cases can be correlated to land-cover vegetation classes. The model is tested in a globally scaled validation exercise involving very recently published reference data and aimed at recognizing the BA class in time series of S2 images. Results are promising as a pure one-class image classification model relying only on MS data reports significantly higher accuracy than those reported for three state-of-the-art BA products currently available, which are obtained through complicated processing stages involving a variety of ancillary inputs and intensive training collection campaigns. The model also outperforms the well-known and largely used OC-SVM, which is also tested in this work for the first time for BA detection at a global scale.

From the theoretical standpoint, the framework proposed in this work opens up for further exploration, and two main directions are envisaged. The first one relates to enforcing the general character of the one-class classification model itself. As a consequence of the general framework described in this work, which is solely based on the mathematical properties of NDIs, its application can potentially encompass a larger variety of different scenarios, provided that the NDIs are suitable descriptors of the classes involved. An extension of this study to other multimodal image domains such as hyperspectral imaging (both for RS and industrial applications) is the path in this direction. The second one aims at exploring possible connections of the proposed model with the field of NNs. At their core, linear NNs are able to represent complex data through the action of neurons each one subdividing the feature space into half-planes through the general parameterized inequality $\sum_i \alpha_i x i < 0$. From direct observation, in our model, we arrived at core inequalities of the type $x_i x_j - x_h x_k < 0$, which can be understood as parameterless versions of a particular case of nonlinear networks that involve nontrivial subsets of the feature space defined by parameterized neurons of the type $\alpha x_i x_j - \beta x_h x_k < 0$. Future studies will involve more insights on the aspects sketched above.

APPENDIX
PROOF OF PROPOSITION 1 AND PSEUDOCODE OF THE CLASSIFICATION ALGORITHM

After performing some simple algebra, we have that

$$\frac{x_i - x_k}{x_i + x_k} - \frac{x_h - x_j}{x_h + x_j} = 2\frac{x_i x_j - x_k x_h}{(x_i + x_k)(x_h + x_j)} \lesseqgtr 0 \tag{18}$$

therefore, because of nonnegativity of the terms, characterization of the sign of NDI differences is equivalent to the characterization of the sign of 2nd-degree binomials

$$x_i x_j - x_k x_h \lesseqgtr 0. \tag{19}$$

Let us prove that all possible nonredundant cases are given by any choice of i, j, k, h with $i \le j$ and $i < k \le h$, with both $k, h \neq j$. First, we can both assume that $i \le j$ and $k \le h$ because multiplication is commutative. We can also assume that $i < k$ as redundant cases are obtained otherwise.

1) *Case $i > k$:* Multiplication of inequality (19) by -1 would reorder terms by fulfilling the assumptions and only changing the sign of the inequality, therefore leaving characterization unaffected.
2) *Case $i = k$:* If $x_i = x_k = a$, then $x_i x_j - x_k x_h = a(x_j - x_h) \lesseqgtr 0$ if and only if $x_j^2 - x_h^2 \lesseqgtr 0$, which is already given by the case $i = j$ and $k = h$.

The same latter argument works to prove that we can also assume $k \neq j$ and $h \neq j$, and the first part of the proof is done. Let us now count how many different choices of i, j, k, and h with $i \le j$ and $i < k \le h$, with both $k, h \neq j$, are possible. Let us fix any $i = 1, \ldots, N-2$, it must be $j \ge i$; therefore, j is chosen $j = i, \ldots, N$. On the one side, when $j = i$, there are $\sum_{t=1}^{N-i} t$ many polynomials of the type ${x_i}^2 - x_k x_h$, with $i < k \le h$ and both $k, h \neq j$. On the other side, for any $j > i$, there are $\sum_{t=1}^{N-i-1} t$ many polynomials of the type $x_i x_j - x_k x_h$, with $i < k \le h$ and both $k, h \neq j$. There are $N - i$ such j; therefore, for the fixed i, we count $\sum_{t=1}^{N-i-1}(N-i)t$ many polynomials as above. For $i, j = N-1$, the only left polynomial is made with $k, h = N-1$. Therefore, the proof is given showing that

$$\left(\sum_{i=1}^{N-2}\sum_{t=1}^{N-i} t + \sum_{i=1}^{N-2}(N-i)\sum_{t=1}^{N-i-1} t\right) + 1 \tag{20}$$

equals $\binom{N}{2} + 3\binom{N}{3} + 3\binom{N}{4} = (N^4 - 2N^3 + 3N^2 - 2N)/8$. In the following, we will make use of three well-known identities for the summation of consecutive nonnegative integers, squares, and cubes.

1) $\sum_{t=1}^{m} t = (m(m+1)/2)$.
2) $\sum_{t=1}^{m} t^2 = (m(m+1)(2m+1)/6)$.
3) $\sum_{t=1}^{m} t^3 = (m^2(m+1)^2/4)$.

Adding and subtracting one more term (i.e., for $t = N - i$) in the last summation of (20), we can group the two sums over index t, simplifying the computation to

$$\sum_{i=1}^{N-2}\left[\left((N-i+1)\sum_{t=1}^{N-i-1} t\right) - (N-i)^2\right] + 1. \tag{21}$$

Applying 1) to the inner summation (with $m = N - i - 1$) and performing multiplications within square brackets, we get

$$\frac{1}{2}\left(\sum_{i=1}^{N-2}(N^3 + N) - (3N^2 + 1)i + 3Ni^2 - i^3\right) + 1. \tag{22}$$

Applying the three formulas for summation of consecutive integers, squares, and cubes for each term in the summation above (indexed with i and with $m = N - 2$), we finally get

$$\frac{1}{2}\left((N^3 + N)(N-2) - (3N^2 + 1)\frac{(N-2)(N-1)}{2} + 3N\frac{(N-2)(N-1)(2N-3)}{6} - \frac{(N-2)^2(N-1)^2}{4}\right) + 1 \tag{23}$$

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
LIST OF ALL POSSIBLE CONFIGURATIONS IN WHICH TWO UP TO FOUR DIFFERENT TERMS CAN BE ARRANGED IN EXPRESSION (19) WITH ITS INDEX CONSTRAINTS

2 terms	3 terms	4 terms
$a^2 < b^2$	$a^2 < bc$	$ab < cd$
	$ab < c^2$	$ac < bd$
	$ac < b^2$	$ad < bc$

Algorithm 1 Pseudocode for the Implementation of Transformation φ in the Classification Case[7]

Require: $\boldsymbol{x} \in \mathbb{R}_+^N$
Ensure: $z = \varphi(\boldsymbol{x})$
$\boldsymbol{d}$ defined as in Section III-B1
$q = 1$
for $i \in \{1, \ldots, N-1\}$ **do**
  for $j \in \{i+1, \ldots, N\}$ **do**
    if $\boldsymbol{d}_q = 0$ **then**
      $z_q = (x_i < x_j)$
    end if
    $q \mathrel{+}= 1$
  end for
  $x_{temp} = x_i x_j$
  for $\ell \in \{1, \ldots, N\} \setminus \{i, j\}$ **do**
    if $\boldsymbol{d}_q = 0$ **then**
      $z_q = (x_{temp} < x_\ell)$
    end if
    $q \mathrel{+}= 1$
  end for
  if $i < N-2$ **then**
    for $k \in \{i+1, \ldots, N-1\} \setminus \{j\}$ **do**
      for $h \in \{k+1, \ldots, N\} \setminus \{j\}$ **do**
        if $\boldsymbol{d}_q = 0$ **then**
          $z_q = (x_{temp} < x_k x_h)$
        end if
        $q \mathrel{+}= 1$
      end for
    end for
  end if
end for

which after some tedious algebra becomes $(N^4 - 2N^3 + 3N^2 - 2N)/8$. Better visualization of the meaning of the binomial terms in expression (11) can be given by listing all possible configurations in which two up to four different terms can be arranged in expression (19) with its index constraints. These cases are listed in Table VIII, and the proof is concluded by noting that there are exactly $\binom{N}{2}$ ways of choosing two different terms, $\binom{N}{3}$ ways of choosing three different terms, and $\binom{N}{4}$ ways of choosing four different terms, among N.

[7]Implementation of the training case follows the same scheme, but it requires eliminating the `if` statements depending on $\boldsymbol{c}_q$.

REFERENCES

[1] S. Plummer, P. Lecomte, and M. Doherty, "The ESA climate change initiative (CCI): A European contribution to the generation of the global climate observing system," *Remote Sens. Environ.*, vol. 203, pp. 2–8, Dec. 2017, doi: 10.1016/j.rse.2017.07.014.

[2] G. Secretariat, *Systematic Observation Requirements for Satellite-Based Products for Climate (2011 Update)*, vol. 154. Copenhagen, Denmark: Global Climate Observing System (GCOS), 2011, pp. 7–18.

[3] I. Z. Gitas, J. San-Miguel-Ayanz, E. Chuvieco, and A. Camia, "Advances in remote sensing and GIS applications in support of forest fire management," *Int. J. Wildland Fire*, vol. 23, no. 5, pp. 603–605, Aug. 2014, doi: 10.1071/WF14117.

[4] K. Thonicke, S. Venevsky, S. Sitch, and W. Cramer, "The role of fire disturbance for global vegetation dynamics: Coupling fire into a dynamic global vegetation model," *Global Ecol. Biogeography*, vol. 10, no. 6, pp. 661–677, Nov. 2001, doi: 10.1046/j.1466-822X.2001.00175.x.

[5] M. O. Andreae, "Emission of trace gases and aerosols from biomass burning—An updated assessment," *Atmos. Chem. Phys.*, vol. 19, no. 13, pp. 8523–8546, Jul. 2019, doi: 10.5194/acp-19-8523-2019.

[6] I. Csiszar et al., "Fire disturbance. Essential climate variables GTOS 68," Food Agricult. Org. United Nations (FAO), Rome, Italy, Tech. Rep., 2009. [Online]. Available: https://lpvs.gsfc.nasa.gov/PDF/TerrestrialECV/T13.pdf

[7] J. M. C. Pereira et al., "Spectral characterisation and discrimination of burnt areas," in *Remote Sensing of Large Wildfires*, E. Chuvieco, Ed. Berlin, Germany: Springer, 1999, pp. 123–138, doi: 10.1007/978-3-642-60164-4_7.

[8] J. M. N. Silva, J. F. C. L. Cadima, J. M. C. Pereira, and J.-M. Grégoire, "Assessing the feasibility of a global model for multi-temporal burned area mapping using SPOT-VEGETATION data," *Int. J. Remote Sens.*, vol. 25, no. 22, pp. 4889–4913, Nov. 2004.

[9] E. Chuvieco, M. P. Martín, and A. Palacios, "Assessment of different spectral indices in the red-near-infrared spectral domain for burned land discrimination," *Int. J. Remote Sens.*, vol. 23, no. 23, pp. 5103–5110, Jan. 2002.

[10] E. Chuvieco et al., "Historical background and current developments for mapping burned area from satellite Earth observation," *Remote Sens. Environ.*, vol. 225, pp. 45–64, May 2019.

[11] D. Riaño, J. A. M. Ruiz, D. Isidoro, and S. L. Ustin, "Global spatial patterns and temporal trends of burned area between 1981 and 2000 using NOAA-NASA pathfinder," *Global Change Biol.*, vol. 13, no. 1, pp. 40–50, Jan. 2007.

[12] R. Ramo and E. Chuvieco, "Developing a random forest algorithm for MODIS global burned area classification," *Remote Sens.*, vol. 9, no. 11, p. 1193, Nov. 2017.

[13] E. Dragozi, I. Gitas, D. Stavrakoudis, and J. Theocharis, "Burned area mapping using support vector machines and the FuzCoC feature selection method on VHR IKONOS imagery," *Remote Sens.*, vol. 6, no. 12, pp. 12005–12036, Dec. 2014.

[14] A. T. Hudak et al., "Sensitivity of Landsat image-derived burn severity indices to immediate post-fire effects," in *Proc. 3rd Int. Fire Ecology Manag. Congr.*, 2006. [Online]. Available: https://www.fs.usda.gov/research/treesearch/26200

[15] D. Fornacca, G. Ren, and W. Xiao, "Evaluating the best spectral indices for the detection of burn scars at several post-fire dates in a mountainous region of Northwest Yunnan, China," *Remote Sens.*, vol. 10, no. 8, p. 1196, Jul. 2018.

[16] E. Roteta, A. Bastarrika, M. Padilla, T. Storm, and E. Chuvieco, "Development of a Sentinel-2 burned area algorithm: Generation of a small fire database for sub-saharan Africa," *Remote Sens. Environ.*, vol. 222, pp. 1–17, Mar. 2019, doi: 10.1016/j.rse.2018.12.011.

[17] M. L. Humber, L. Boschetti, L. Giglio, and C. O. Justice, "Spatial and temporal intercomparison of four global burned area products," *Int. J. Digit. Earth*, vol. 12, no. 4, pp. 460–484, Apr. 2019.

[18] M. Zanetti et al., "A system for burned area detection on multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404315.

[19] R. Meng and F. Zhao, "Remote sensing of fire effects. A review for recent advances in burned area and burn severity mapping," in *Remote Sensing of Hydrometeorological Hazards*, G. P. Petropulos and T. Islam, Eds. Boca Raton, FL, USA: Taylor & Francis, 2017.

[20] A. T. Hudak et al., "The relationship of multispectral satellite imagery to immediate fire effects," *Fire Ecol.*, vol. 3, no. 1, pp. 64–90, Jun. 2007, doi: 10.4996/fireecology.0301064.

[21] M. M. Moya and D. R. Hush, "Network constraints and multi-objective optimization for one-class classification," *Neural Netw.*, vol. 9, no. 3, pp. 463–474, Apr. 1996.

[22] W. Li, Q. Guo, and C. Elkan, "A positive and unlabeled learning algorithm for one-class classification of remote-sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 717–725, Feb. 2011.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.

[23] N. Seliya, A. A. Zadeh, and T. M. Khoshgoftaar, "A literature review on one-class classification and its potential applications in big data," *J. Big Data*, vol. 8, no. 1, pp. 1–31, Sep. 2021.

[24] P. Baldi and K. Hornik, "Neural networks and principal component analysis: Learning from examples without local minima," *Neural Netw.*, vol. 2, no. 1, pp. 53–58, Jan. 1989.

[25] V. Barnett and T. Lewis, *Outliers in Statistical Data* (Wiley Series in Probability and Mathematical Statistics). Hoboken, NJ, USA: Wiley, 1984.

[26] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, Jul. 2001.

[27] D. M. J. Tax and R. P. W. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, Jan. 2004.

[28] N. Koutsias, G. Mallinis, and M. Karteris, "A forward/backward principal component analysis of Landsat-7 ETM+ data to enhance the spectral signal of burnt surfaces," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 1, pp. 37–46, Jan. 2009.

[29] N. Koutsias and M. Karteris, "Logistic regression modelling of multitemporal thematic mapper data for burned area mapping," *Int. J. Remote Sens.*, vol. 19, no. 18, pp. 3499–3514, Dec. 1998, doi: 10.1080/014311698213777.

[30] A. Pereira et al., "Burned area mapping in the Brazilian savanna using a one-class support vector machine trained by active fires," *Remote Sens.*, vol. 9, no. 11, p. 1161, Nov. 2017, doi: 10.3390/rs9111161.

[31] M. Zanetti and L. Bruzzone, "A theoretical framework for change detection based on a compound multiclass statistical model of the difference image," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1129–1143, Feb. 2018.

[32] R. B. Myneni, F. G. Hall, P. J. Sellers, and A. L. Marshak, "The interpretation of spectral vegetation indexes," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 2, pp. 481–486, Mar. 1995, doi: 10.1109/TGRS.1995.8746029.

[33] D. Murray, *Biophysical Ecology* (Springer Advanced Texts in Life Sciences). Minneapolis, MN, USA: Univ. Minnesota, 1980.

[34] F. Kriegler, W. Malila, R. Nalepka, and W. Richardson, "Preprocessing transformations and their effects on multispectral recognition," *Remote Sens. Environ., VI*, p. 97, 1969. [Online]. Available: https://ui.adsabs.harvard.edu/abs/1969rse..conf...97K/abstract

[35] J. M. C. Pereira, "A comparative evaluation of NOAA/AVHRR vegetation indexes for burned surface detection and mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 1, pp. 217–226, 1999.

[36] C. H. Key and N. C. Benson, "Landscape assessment: Ground measure of severity, the composite burn index; and remote sensing of severity, the normalized burn ratio," FIREMON, USDA Forest Service, Ogden, UT, USA, RMRS-GTR-164-CD Tech. Rep., 164, pp. 1–55, 2006.

[37] M. J. L. García and V. Caselles, "Mapping burns and natural reforestation using thematic mapper data," *Geocarto Int.*, vol. 6, no. 1, pp. 31–37, Mar. 1991.

[38] J. Pereira, E. Chuvieco, A. Beaudoin, and N. Desbois, "Remote sensing of burned areas: A review," in *A Review of Remote Sensing Methods for the Study of Large Wildland Fires*, E. Chuvieco, Ed. Univ. de Alcala, 1997, pp. 127–185.

[39] S. Escuin, R. Navarro, and P. Fernández, "Fire severity assessment by using NBR (normalized burn ratio) and NDVI (normalized difference vegetation index) derived from LANDSAT TM/ETM images," *Int. J. Remote Sens.*, vol. 29, no. 4, pp. 1053–1073, Feb. 2008, doi: 10.1080/01431160701281072.

[40] P. R. Robichaud, S. A. Lewis, D. Y. M. Laes, A. T. Hudak, R. F. Kokaly, and J. A. Zamudio, "Postfire soil burn severity mapping with hyperspectral image unmixing," *Remote Sens. Environ.*, vol. 108, no. 4, pp. 467–480, Jun. 2007.

[41] D. P. Roy, L. Boschetti, and S. N. Trigg, "Remote sensing of fire severity: Assessing the performance of the normalized burn ratio," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 112–116, Jan. 2006, doi: 10.1109/LGRS.2005.858485.

[42] K. A. Hammill and R. A. Bradstock, "Remote sensing of fire severity in the Blue Mountains: Influence of vegetation type and inferring fire intensity," *Int. J. Wildland Fire*, vol. 15, no. 2, pp. 213–226, Jun. 2006, doi: 10.1071/WF05051.

[43] L. Telesca and R. Lasaponara, "Fire-induced variability in satellite SPOT-VGT NDVI vegetational data," *Int. J. Remote Sens.*, vol. 27, no. 14, pp. 3087–3095, Jul. 2006.

[44] P. R. Coppin and M. E. Bauer, "Digital change detection in forest ecosystems with remote sensing imagery," *Remote Sens. Rev.*, vol. 13, nos. 3–4, pp. 207–234, Apr. 1996.

[45] C. Key and N. Benson, "The normalized burn ratio, a Landsat TM radiometric index of burn severity incorporating multi-temporal differencing," U.S. Geol. Survey, Reston, VA, USA, Tech. Rep., 1999. [Online]. Available: https://www.frames.gov/catalog/3356

[46] J. D. Miller and A. E. Thode, "Quantifying burn severity in a heterogeneous landscape with a relative version of the delta normalized burn ratio (dNBR)," *Remote Sens. Environ.*, vol. 109, no. 1, pp. 66–80, Jul. 2007, doi: 10.1016/j.rse.2006.12.006.

[47] J. García-Lázaro, J. Moreno-Ruiz, D. Riaño, and M. Arbelo, "Estimation of burned area in the northeastern Siberian boreal forest from a long-term data record (LTDR) 1982–2015 time series," *Remote Sens.*, vol. 10, no. 6, p. 940, Jun. 2018.

[48] D. A. Bini, B. Meini, and F. Poloni, "On the solution of a quadratic vector equation arising in Markovian binary trees," *Numer. Linear Algebra Appl.*, vol. 18, no. 6, pp. 981–991, Nov. 2011.

[49] M. Franquesa et al., "Development of a standard database of reference sites for validating global burned area products," *Earth Syst. Sci. Data*, vol. 12, no. 4, pp. 3229–3246, Dec. 2020.

[50] M. Franquesa, J. Lizundia-Loiola, S. V. Stehman, and E. Chuvieco, "Using long temporal reference units to assess the spatial accuracy of global satellite-derived burned area products," *Remote Sens. Environ.*, vol. 269, Feb. 2022, Art. no. 112823.

[51] E. Dinerstein et al., "An ecoregion-based approach to protecting half the terrestrial realm," *BioScience*, vol. 67, no. 6, pp. 534–545, Jun. 2017.

[52] B. Schölkopf, A. J. Smola, R. C. Williamson, and P. L. Bartlett, "New support vector algorithms," *Neural Comput.*, vol. 12, no. 5, pp. 1207–1245, May 2000.

[53] Y. T. Solano-Correa, D. Carcereri, F. Bovolo, and L. Bruzzone, "Identification of non-photosynthetic vegetation areas in Sentinel-2 satellite image time series," *Proc. SPIE*, vol. 11155, pp. 304–312, Oct. 2019.



Massimo Zanetti (Member, IEEE) received the M.S. degree (summa cum laude) in mathematics from the Department of Mathematics and Computer Science, University of Ferrara, Ferrara, Italy, in 2013, and the Ph.D. degree (cum laude) in communication and information technologies from the Department of Information Engineering and Computer Science, University of Trento, Trento, Italy, in 2017.

He is currently a Researcher with the Remote Sensing for Digital Earth Unit, Fondazione Bruno Kessler (FBK), Trento. His research interests include multispectral/hyperspectral image change detection, time series analysis and variational methods for segmentation, and numerical optimization methods for image processing problems.

Dr. Zanetti was a recipient of the prizes for the GTTI Ph.D. Award in 2017 for the Best Ph.D. Thesis in Telecommunications and Information Technologies, in Italy, awarded by the Gruppo Telecomunicazioni e Tecnologie dell'Informazione, and the Best Ph.D. Thesis Award in 2018 for the Best Ph.D. Thesis presented in the area of geoscience and remote sensing, in Italy, awarded by the Italy Chapter of the IEEE Geoscience and Remote Sensing Society. He is a reviewer for international journals, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Create Net- Fondazione Bruno Kessler. Downloaded on August 17,2023 at 07:44:40 UTC from IEEE Xplore. Restrictions apply.